

SQL Retail Database Practice

Overview

This project demonstrates SQL fundamentals using a sample Retail Database. It covers database creation, table design, relationships, data manipulation, filtering, and aggregate functions. The project is intended for beginners learning SQL and for interview preparation.

Database Structure

The database consists of three related tables:

- **Salespeople** – Stores salesperson details.
- **Customer** – Stores customer information and references the salesperson.
- **Orders** – Stores customer orders and references the customer.

Concepts Covered

- Database Creation (DDL)
- Table Creation
- Primary Keys
- Foreign Keys
- Data Types
- Insert Single & Multiple Records
- SELECT Statements
- WHERE Clause
- UPDATE Statements
- DELETE Statements
- AND, OR Operators
- IN Operator
- BETWEEN Operator
- DISTINCT
- Aggregate Functions (COUNT)
- Table Information using `sp_help`

SQL Operations Included

- Creating a database
- Creating parent and child tables
- Establishing table relationships
- Inserting records
- Retrieving records
- Updating records
- Deleting records

- Filtering data
- Counting records
- Finding unique values

Learning Outcome

This project helped strengthen my understanding of relational databases, SQL syntax, data manipulation, filtering techniques, and database relationships using primary and foreign keys. It also serves as practice for Data Analyst and SQL interview preparation.